

Database Systems

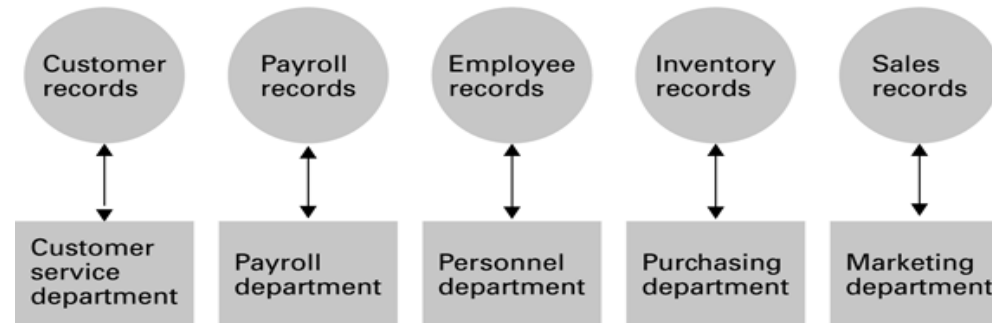
# **CSCI 1101 Introduction to Computer Science**

# Database definition

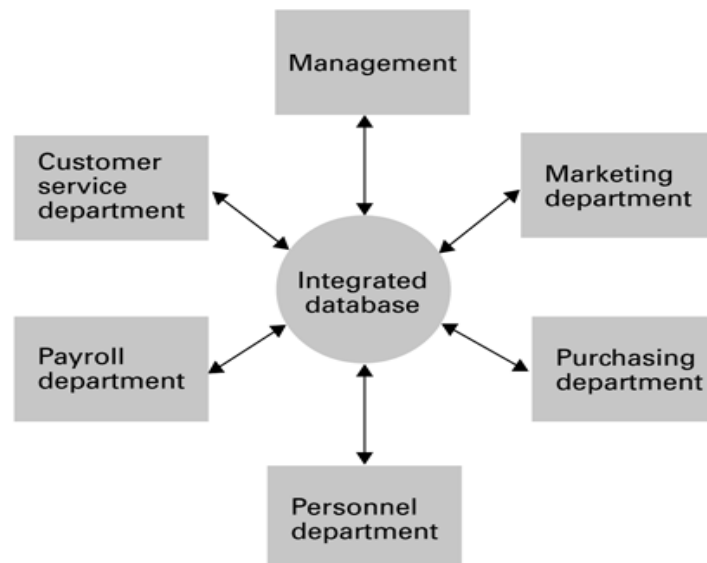
- A usually large collection of data organized especially for rapid search and retrieval (as by a computer) (Merriam-Webster)

# A file versus a database organization

a. File-oriented information system



b. Database-oriented information system



# Advantages of database systems over file system

The integration of the database makes it:

- Structured (easy to modify)
- Easy to access and maintain
- Easy to distribute
- Provide minimal redundancy of data
- Provide better data integrity (accuracy and consistency of data)
- Safer
- Allow for different types of access to data to different type of users

# Components of a Database System

- **Actual (physical) database**

The collection of files that contain the data

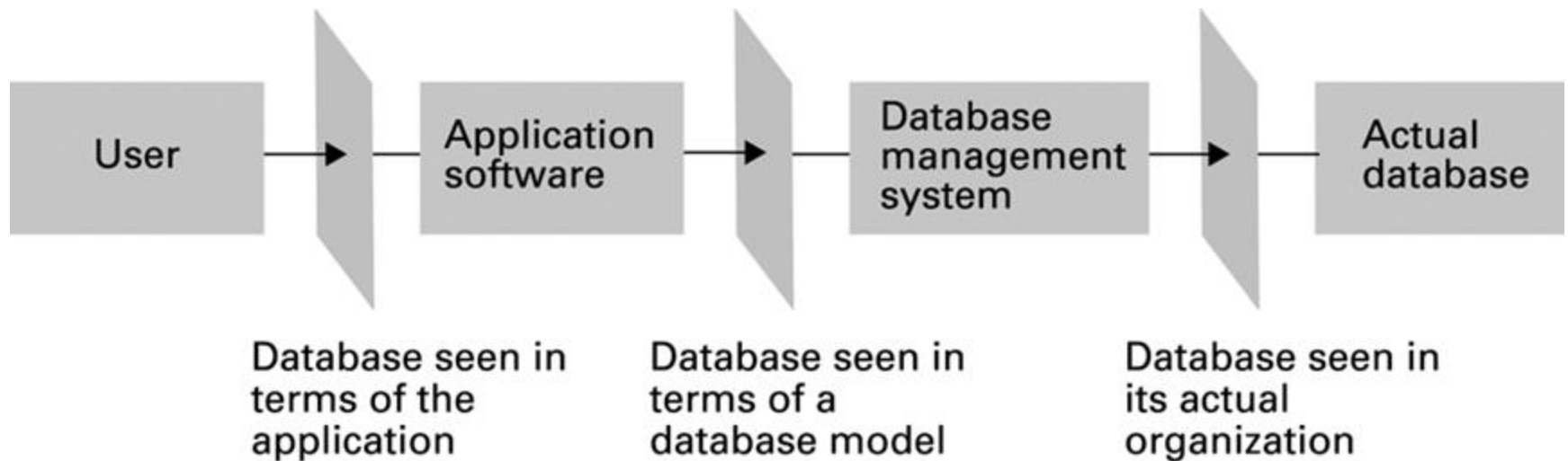
- **Database engine**

The software layer that supports access to and modification of the database contents

- **Database Management System (DBMS)**

The software layer that manipulates a database in response to requests from applications

# The conceptual layers of a database implementation



# Database Management Systems

The separation of the application software layer from the DBMS layer facilitates:

- **Data independence:** The ability to change the organization of a database without changing the application software that uses it
- **Controlled access to the database:** the DBMS can enforce different restrictions to different users of the system
- **Distributed Database:** A database stored on multiple machines
  - DBMS will mask this organizational detail from its users

# Database Models

- **Database model:** A conceptual view of a database
  - **Relational database model:** A system in which the data items and the relationships among them are organized into **tables**
  - **Object-oriented database model:** A system in which information is represented in the form of objects (the data itself and the operations that can be performed on them)

# Relational Database Model

- **Relation:** A rectangular **table** that represents a group of data items with common attributes
  - **Field:** A column in the table. Represents an attribute.
  - **Record:** A row in the table. Represents an individual from the group.

# A relation containing employee information

| Empl Id | Name            | Address           | SSN       |
|---------|-----------------|-------------------|-----------|
| 25X15   | Joe E. Baker    | 33 Nowhere St.    | 111223333 |
| 34Y70   | Cheryl H. Clark | 563 Downtown Ave. | 999009999 |
| 23Y34   | G. Jerry Smith  | 1555 Circle Dr.   | 111005555 |
| •       | •               | •                 | •         |
| •       | •               | •                 | •         |
| •       | •               | •                 | •         |

**Relation:** Represents a group of employees from a company

**Field:** Represents an attribute (name, address, etc.) of the employees

**Record:** Represents an employee in the group

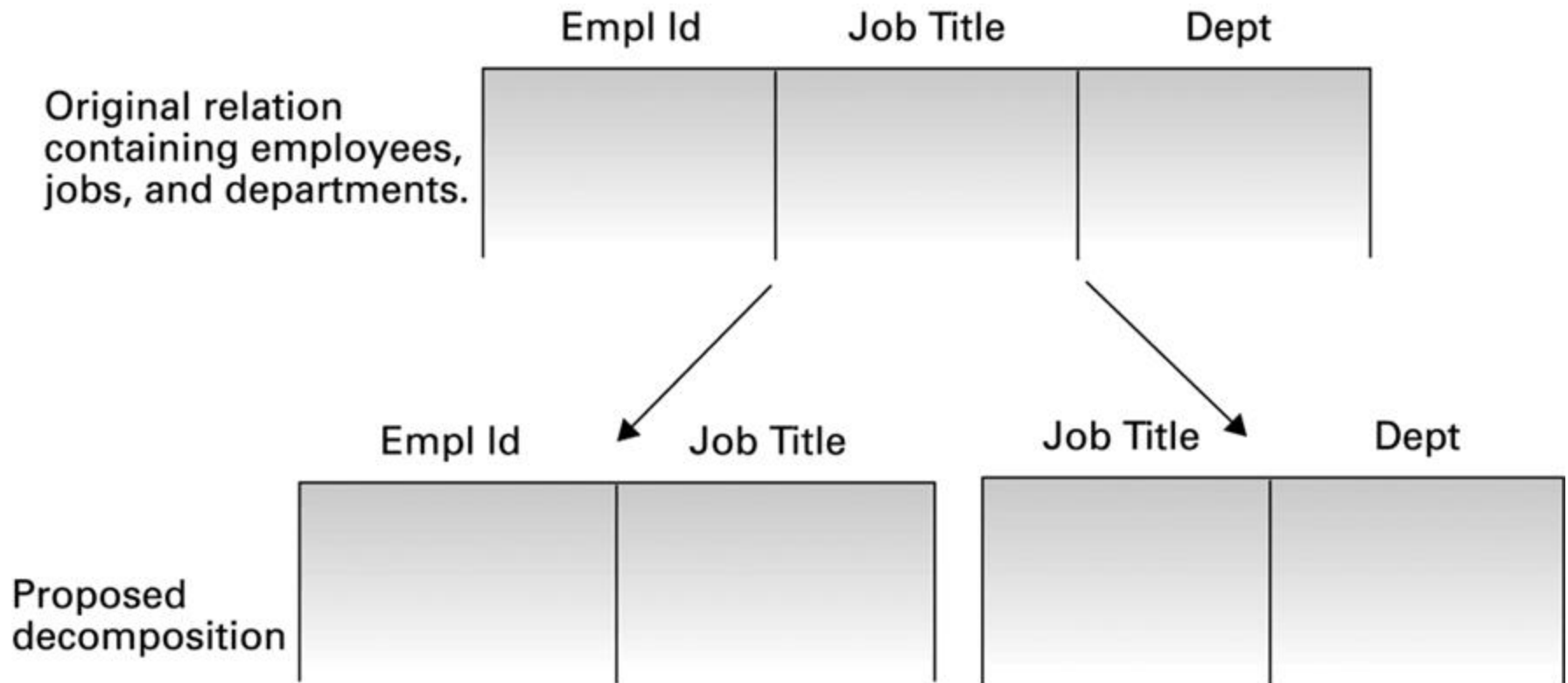
# Relational Design

- Avoid multiple concepts within one relation
  - Can lead to redundant data
  - Deleting a record could also delete necessary but unrelated information

# Improving a Relational Design

- **Decomposition:** Dividing the columns of a relation into two or more relations, duplicating those columns necessary to maintain relationships
  - **Lossless** or **nonloss** decomposition: A “correct” decomposition that does not lose any information

# A relation and a proposed decomposition



# A relation containing redundancy

| Empl Id | Name            | Address           | SSN       | Job Id | Job Title     | Skill Code | Dept       | Start Date | Term Date |
|---------|-----------------|-------------------|-----------|--------|---------------|------------|------------|------------|-----------|
| 25X15   | Joe E. Baker    | 33 Nowhere St.    | 111223333 | F5     | Floor manager | FM3        | Sales      | 9-1-2007   | 9-30-2008 |
| 25X15   | Joe E. Baker    | 33 Nowhere St.    | 111223333 | D7     | Dept. head    | K2         | Sales      | 10-1-2008  | *         |
| 34Y70   | Cheryl H. Clark | 563 Downtown Ave. | 999009999 | F5     | Floor manager | FM3        | Sales      | 10-1-2007  | *         |
| 23Y34   | G. Jerry Smith  | 1555 Circle Dr.   | 111005555 | S25X   | Secretary     | T5         | Personnel  | 3-1-1999   | 4-30-2006 |
| 23Y34   | G. Jerry Smith  | 1555 Circle Dr.   | 111005555 | S26Z   | Secretary     | T6         | Accounting | 5-1-2006   | *         |
| .       | .               | .                 | .         | .      | .             | .          | .          | .          | .         |
| .       | .               | .                 | .         | .      | .             | .          | .          | .          | .         |
| .       | .               | .                 | .         | .      | .             | .          | .          | .          | .         |

# An employee database consisting of three relations

**EMPLOYEE relation**

| Empl Id | Name            | Address           | SSN       |
|---------|-----------------|-------------------|-----------|
| 25X15   | Joe E. Baker    | 33 Nowhere St.    | 111223333 |
| 34Y70   | Cheryl H. Clark | 563 Downtown Ave. | 999009999 |
| 23Y34   | G. Jerry Smith  | 1555 Circle Dr.   | 111005555 |
|         |                 |                   |           |

**JOB relation**

| Job Id | JobTitle      | Skill Code | Dept       |
|--------|---------------|------------|------------|
| S25X   | Secretary     | T5         | Personnel  |
| S26Z   | Secretary     | T6         | Accounting |
| F5     | Floor manager | FM3        | Sales      |
| .      | .             | .          | .          |
| .      | .             | .          | .          |
| .      | .             | .          | .          |

**ASSIGNMENT relation**

| Empl Id | Job Id | Start Date | Term Date |
|---------|--------|------------|-----------|
| 23Y34   | S25X   | 3-1-1999   | 4-30-2006 |
| 34Y70   | F5     | 10-1-2007  | *         |
| 23Y34   | S26Z   | 5-1-2006   | *         |
| .       | .      | .          | .         |
| .       | .      | .          | .         |
| .       | .      | .          | .         |

# Finding the departments in which employee 23Y34 has worked

**EMPLOYEE relation**

| Empl Id | Name            | Address           | SSN       |
|---------|-----------------|-------------------|-----------|
| 25X15   | Joe E. Baker    | 33 Nowhere St.    | 111223333 |
| 34Y70   | Cheryl H. Clark | 563 Downtown Ave. | 999009999 |
| 23Y34   | G. Jerry Smith  | 1555 Circle Dr.   | 111005555 |
| .       | .               | .                 | .         |
| .       | .               | .                 | .         |
| .       | .               | .                 | .         |

**JOB relation**

| Job Id | Job Title     | Skill Code | Dept       |
|--------|---------------|------------|------------|
| S25X   | Secretary     | T5         | Personnel  |
| S26Z   | Secretary     | T6         | Accounting |
| F5     | Floor manager | FM3        | Sales      |
| .      | .             | .          | .          |
| .      | .             | .          | .          |
| .      | .             | .          | .          |

are contained  
in the personnel  
and accounting  
departments.

**ASSIGNMENT relation**

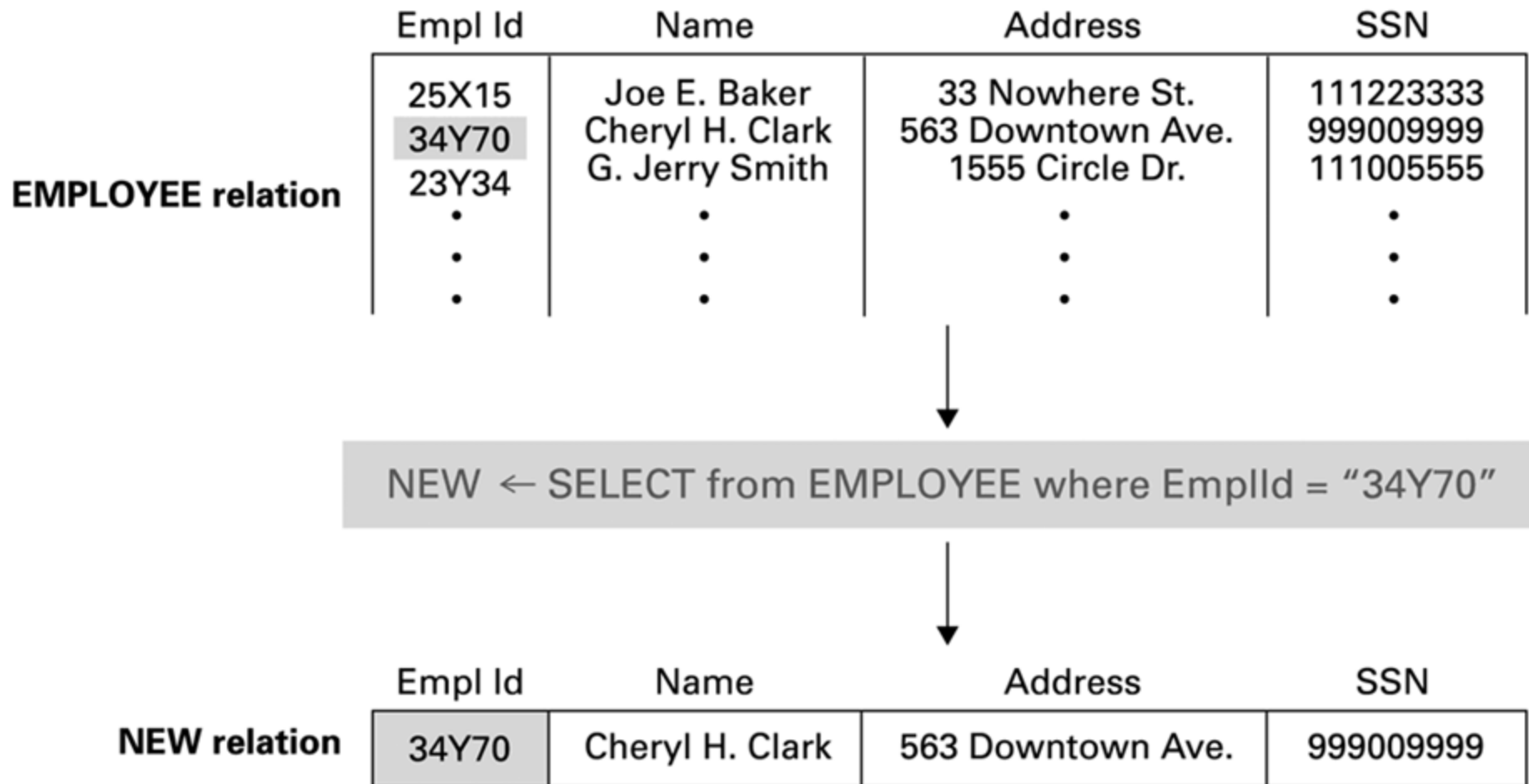
| Empl Id | Job Id | Start Date | Term Date |
|---------|--------|------------|-----------|
| 23Y34   | S25X   | 3-1-1999   | 4-30-2006 |
| 34Y70   | F5     | 10-1-2007  | *         |
| 23Y34   | S26Z   | 5-1-2006   | *         |
| .       | .      | .          | .         |
| .       | .      | .          | .         |
| .       | .      | .          | .         |

The jobs held  
by employee  
23Y34

# Relational Operations

- **Select:** Choose rows
- **Project:** Choose columns
- **Join:** Assemble information from two or more relations

# The SELECT operation



# The PROJECT operation

| EMPLOYEE relation | Empl Id | Name            | Address           | SSN       |
|-------------------|---------|-----------------|-------------------|-----------|
|                   | 25X15   | Joe E. Baker    | 33 Nowhere St.    | 111223333 |
|                   | 24Y70   | Cheryl H. Clark | 563 Downtown Ave. | 999009999 |
|                   | 23Y34   | G. Jerry Smith  | 1555 Circle Dr.   | 111005555 |
|                   | ⋮       | ⋮               | ⋮                 | ⋮         |

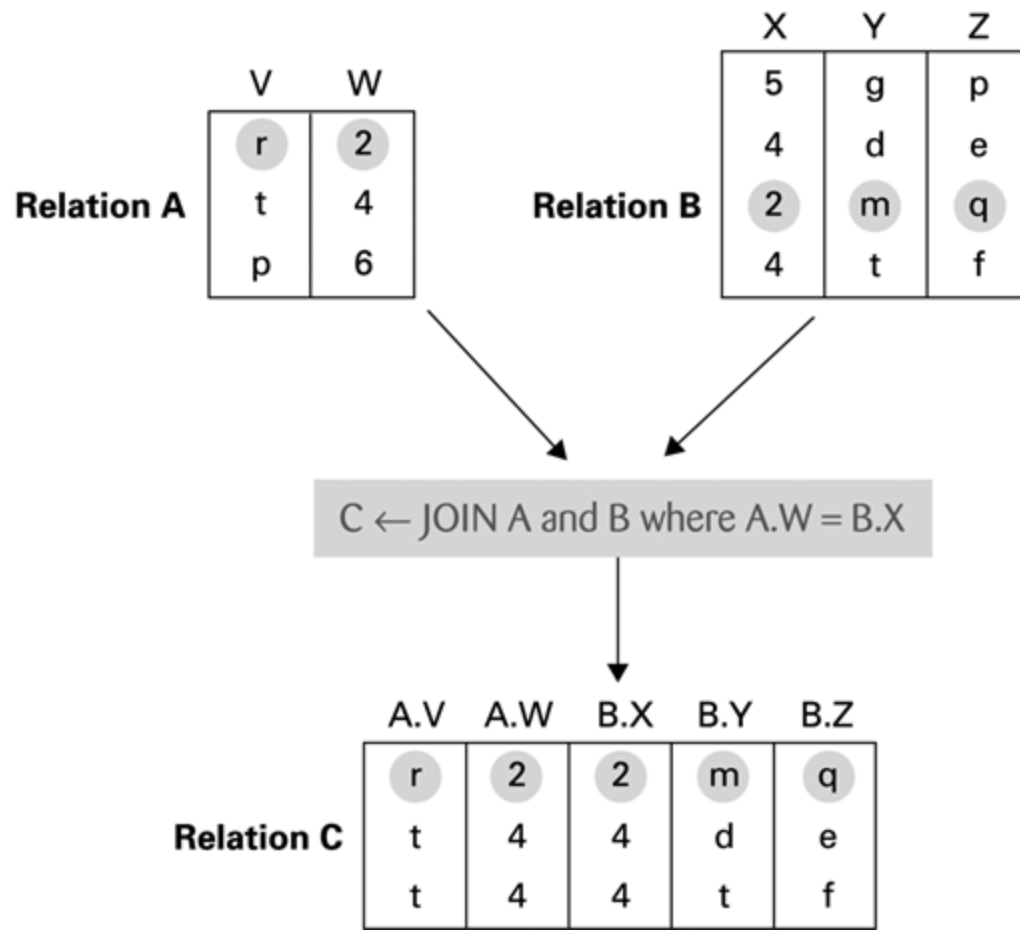


MAIL ← PROJECT Name, Address from EMPLOYEE

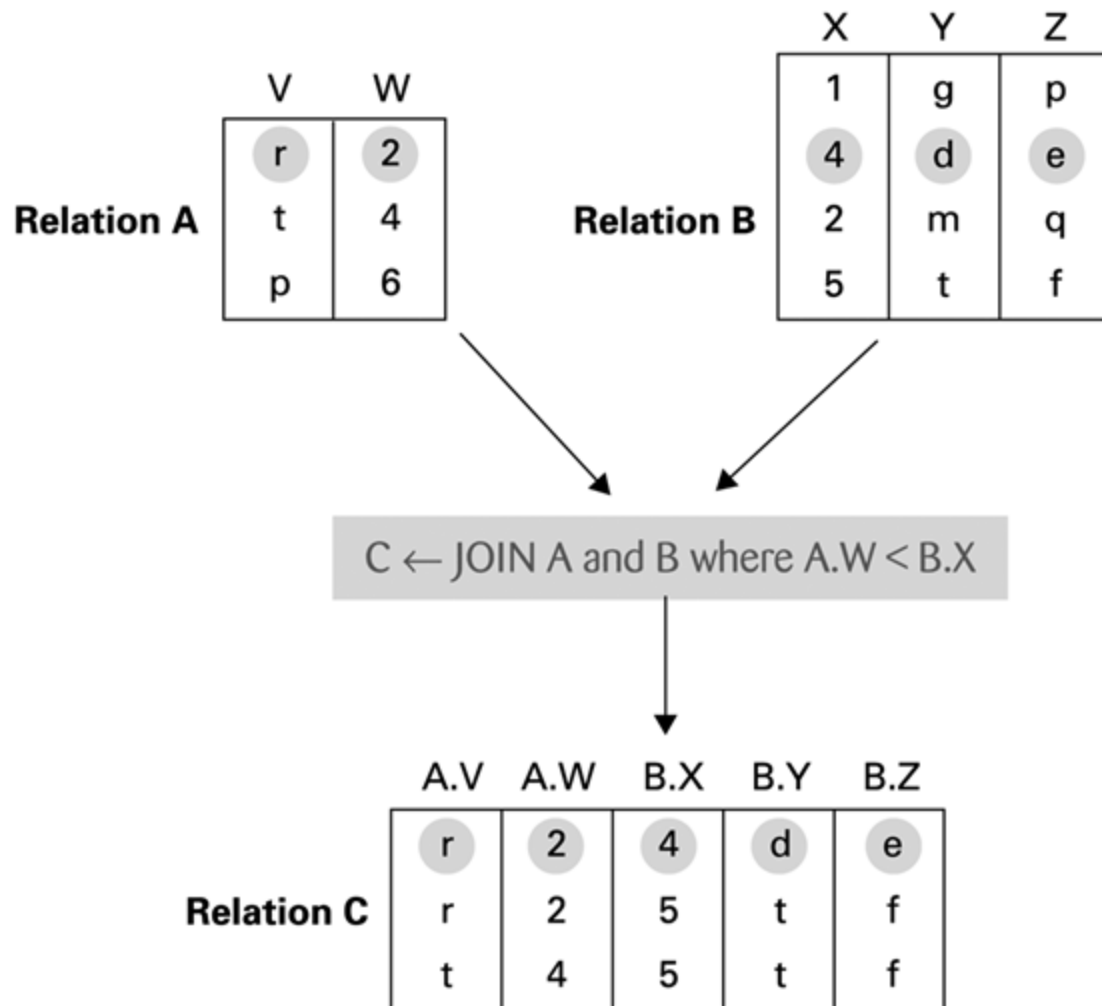


| MAIL relation | Name            | Address           |
|---------------|-----------------|-------------------|
|               | Joe E. Baker    | 33 Nowhere St.    |
|               | Cheryl H. Clark | 563 Downtown Ave. |
|               | G. Jerry Smith  | 1555 Circle Dr.   |
|               | ⋮               | ⋮                 |

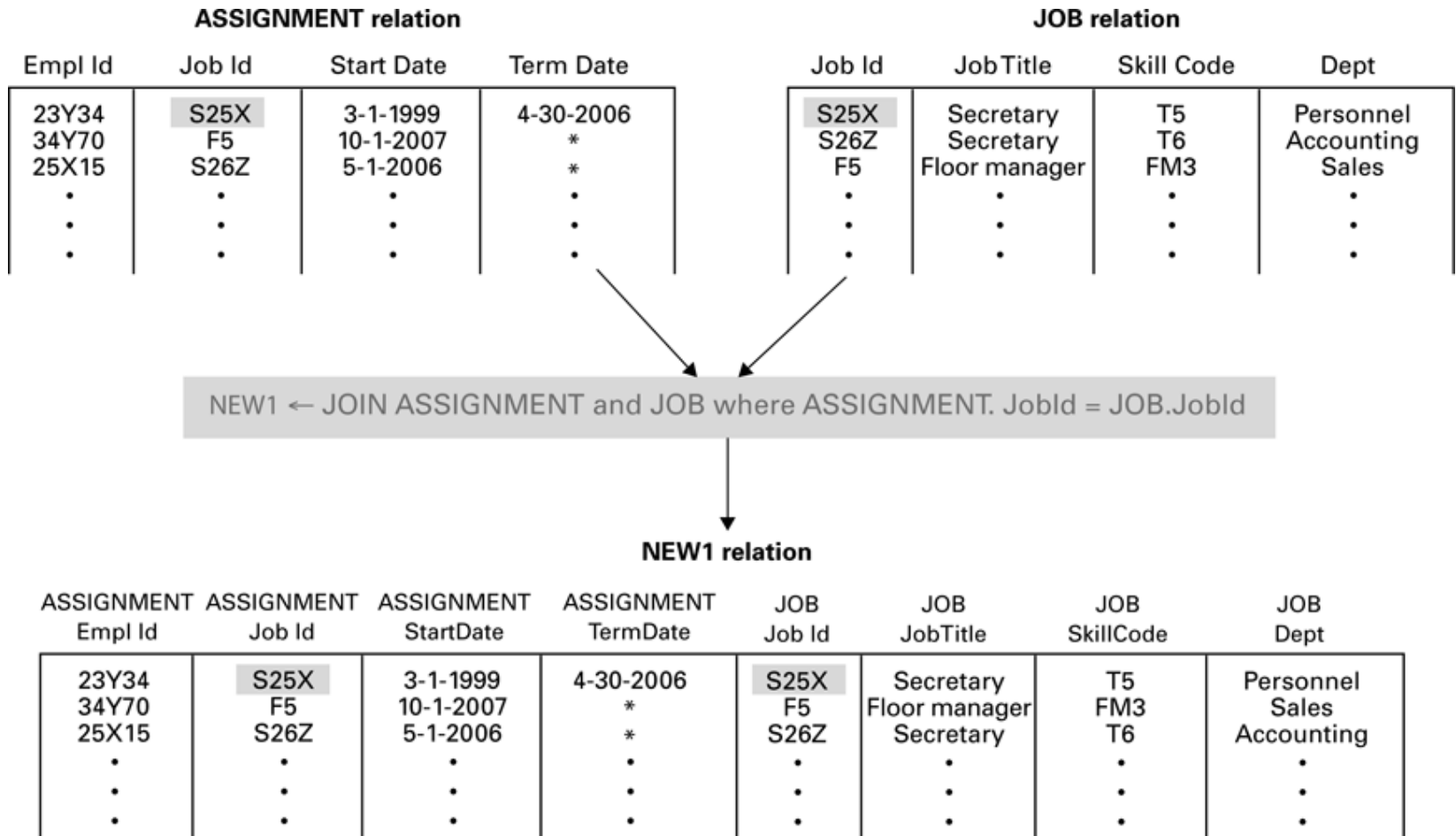
# The JOIN operation



# Another example of the JOIN operation



# An application of the JOIN operation



# Structured Query Language (SQL)

- Operations to manipulate records
  - insert
  - update
  - delete
  - select

# SQL Examples

- ```
select EmplId, Dept
from ASSIGNMENT, JOB
where ASSIGNMENT.JobId = JOB.JobId
and ASSIGNMENT.TermData = "*"

```
- ```
insert into EMPLOYEE
values ('43212', 'Sue A. Burt',
      '33 Fair St.', '444661111')

```

# SQL Examples (continued)

- `delete from EMPLOYEE  
where Name = 'G. Jerry Smith'`
- `update EMPLOYEE  
set Address = '1812 Napoleon Ave.'  
where Name = 'Joe E. Baker'`

# Maintaining Database Integrity

- **Transaction:** A sequence of operations that must all happen together
  - Example: transferring money between bank accounts
- **Transaction log:** A non-volatile record of each transaction's activities, built before the transaction is allowed to execute
  - **Commit point:** The point at which a transaction has been recorded in the log
  - **Roll-back:** The process of undoing a transaction

# Maintaining Database Integrity

- **Transaction:** transferring money between bank accounts
  1. Enter in log: account A balance is decremented
  2. Decrement balance of account A
  3. Enter in log: account B balance is incremented
  4. Increment balance of account B
- **Commit point:** Achieved after steps 1 through 4 have been executed
- **Roll-back:** If something goes wrong and the transaction is not finished, by analyzing the log it is possible to know what needs to be rolled back and perform the transaction again

# Maintaining database integrity (continued)

## Simultaneous access problems

- **Incorrect summary case**
  - The owner of an account checks balance of an account (\$100) and extracts \$80
  - Before the balance is updated, another owner of the account checks the balance (\$100) and extracts \$50
  - The owners of the account extract \$130 from an account with only \$100 !!!

# Maintaining database integrity (continued)

Solution to simultaneous access problems

- **Locking** = preventing others from accessing data being used by a transaction
  - **Shared** lock: used when reading data
  - **Exclusive** lock: used when altering data